

## Homework 2, Math 701 – Frank Thorne (thorne@math.sc.edu)

**Instructions:** You are welcome and encouraged to collaborate, but please write up your own solutions.

Due Wednesday, October 7.

1. (a) Prove, as described in class, that the group  $\mathrm{SL}_2(\mathbb{Z})$  acts on the complex upper half plane

$$\mathbb{H} := \{z \in \mathbb{C} : \Im(z) > 0\}.$$

- (b) Compute the stabilizers of  $i$ ,  $\frac{-1+\sqrt{3}i}{2}$ , and  $2i$ : compute them as sets, and determine familiar groups to which they are isomorphic.

2. Let  $V$  be a finite dimensional vector space with a fixed basis, and write  $V^* = \mathrm{Hom}(V, F)$  for its dual space. Prove that

$$(g \circ f)(v) = f(gv)$$

does not define a group action of  $\mathrm{GL}(V)$  on  $V^*$ , but that either of the equations

$$(g \circ f)(v) = f(g^{-1}v)$$

or

$$(g \circ f)(v) = f(g^T v)$$

do.

3. Let  $G$  be the group of rigid (orientation preserving) motions of a cube.

- (a) The position of a cube is determined by where each of its faces are. (You may accept this as ‘geometrically obvious’). Explain why this yields an injective homomorphism  $G \rightarrow \mathrm{Sym}(6)$ .
- (b)  $G$  also acts on the three-element set of *pairs* of opposite faces. Prove that the resulting homomorphism  $G \rightarrow \mathrm{Sym}(3)$  is *not* injective.
- (c) By describing yet another action of  $G$  on (... something associated to the cube ...!), prove that the group of rigid motions of a cube is isomorphic to  $\mathrm{Sym}(4)$ .
- (d) Combining the last two problems yields a surjective homomorphism  $\mathrm{Sym}(4) \rightarrow \mathrm{Sym}(3)$ . Describe it explicitly and compute its kernel.

4. (a) Determine all the conjugacy classes of  $S_6$ , together with their sizes and the orders of the stabilizers of any element.
- (b) For each conjugacy class, pick a representative element and compute its stabilizer explicitly as a subgroup of  $S_6$ .
- (c) Do both of the above for  $A_6$  in place of  $S_6$ .

5. Prove that  $\mathrm{Aut}(D_4) \simeq D_4$ , and describe an isomorphism explicitly.